

JIZTOM KAVALAKKATT FRANCIS, PhD

Principal / Senior Embedded Systems Engineer · Test Infrastructure · Safety-Critical & Autonomous Systems
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PROFESSIONAL SUMMARY

Embedded systems engineer with 7+ years building reliable firmware and the software/system test infrastructure that validates it on complex, hardware-based platforms. Deep, hands-on track record on safety-significant field vehicles: designed CAN/J1939 data-extraction and characterization systems for agricultural heavy equipment, integrated and calibrated perception stacks (LiDAR, radar, stereo cameras, GPS/GNSS, encoders, IMU) onto embedded controllers, and built automated, HIL-style test rigs for components and integrated systems. Strong in Modern C++ and C for embedded targets, Python for test automation and tooling, and the full development lifecycle on Linux — Git, CI/CD, dev containers, and root-cause analysis from log and bus data. Comfortable at the hardware/software boundary with oscilloscopes, DAQs, signal generators, logic analyzers, and CAN/network analyzers, and accustomed to leading and mentoring small engineering teams.

TECHNICAL SKILLS

Languages: C, Modern C++ (C++17), Python (test automation, tooling, data); MATLAB, LabVIEW, Assembly basics

Software & System Test Infrastructure: automated test rig design, HIL-style component & integrated-system tests, automated characterization & calibration, Test-Driven Development (GoogleTest), DOE methodology (Minitab), CI/CD for firmware builds, log-data triage & root-cause analysis

Perception & Safety-Significant Hardware: LiDAR, radar, stereo/mono cameras, GPS/GNSS, encoders, IMU, analog front ends; sensor fusion, time synchronization, calibration; high-voltage / interlock-style protected front ends

Buses & Networking: CAN / J1939 (extensive), UART, SPI, I2C, GPIO, ADC/DAC, Timers/PWM, BLE, RS232, JTAG/SWD; Ethernet, TCP/UDP, DHCP/DNS; Wireshark & CAN-bus analyzers; REST APIs

Platforms & OS: ARM Cortex-M (STM32, NXP), ESP32 / ESP32-C6 (RISC-V), Arduino/AVR, Raspberry Pi; embedded Linux (Ubuntu, RedHat)

Tooling & DevOps: Git, CI/CD pipelines, Docker / development containers, Make/CMake, GCC/ARM toolchains, GDB, Segger J-Link; AI-assisted development workflows

Robotics & Real-Time: ROS (Iowa State Robotics Club), point-cloud processing (PCL-style), OpenCV; FreeRTOS-style & round-robin scheduling, rate-monotonic/EDF reasoning, ISRs, DMA, concurrency, hard real-time timing

Functional Safety: ISO 15118 EV-charging driver delivered; working familiarity with ISO 26262 / IEC 61508 principles

Adjacent & ML/AI: Deep learning (PyTorch, TensorFlow) for edge inference and production pipelines; NumPy, Pandas, Scikit-learn; RAG/LLM tooling; FastAPI, Docker, Kubernetes, CUDA; AWS, Azure, GCP; SQL/ETL for sensor telemetry; n8n; ESPHome, Home Assistant, QuestDB, Grafana

PROFESSIONAL EXPERIENCE

Graduate Research Engineer — Digital Agriculture (DigitalAg Lab) · Iowa State University
Jan 2019 – Dec 2024

Ames, IA · Firmware, perception integration, and test infrastructure for field-deployed agricultural heavy equipment (tractors, combines, sprayers).

- Designed, implemented, and delivered test solutions for embedded components and integrated systems — automated, HIL-style rigs that exercised firmware, CAN traffic, and sensor inputs for repeatable validation against acquisition and timing targets.
- Developed automated characterization and calibration routines for perception sensors and analog chains (LiDAR, radar, stereo/mono cameras, GPS/GNSS, ADC front ends), handling time synchronization and fusion for in-field vehicle testing.
- Built CAN / J1939 data-extraction systems that decoded vehicle bus and ECU telemetry into clean, structured datasets for downstream analytics and ML pipelines.

- Designed and deployed a custom GPS + CAN field tagger end-to-end — schematic, transceiver selection, GNSS-over-UART integration, C firmware, and on-vehicle validation — broadcasting time-synchronized georeferenced events onto an equipment CAN network.
- Triaged field and integration issues via root-cause analysis from log, bus, and sensor data; validated bus timing, signal integrity, and I2C/SPI/ADC peripherals on oscilloscopes, logic analyzers, and CAN-bus analyzers during board bring-up.
- Wrote C/C++ peripheral drivers (UART, SPI, I2C, ADC, GPIO, CAN) under deterministic schedulers; shipped Linux firmware improvements that raised field acquisition efficiency ~30% via optimized I/O scheduling and interrupt-driven reads.
- Selected and lifecycle-qualified components (CAN transceivers, regulators, GNSS modules, sensors) and authored design specs, schematics, BOMs, and test reports supporting deployment across multiple research sites.

Lead Developer, Real-Time DSP / C++ · Make Believe Studios · Remote Dec 2025 – Present

- Ship production C++ audio plugins (JUCE) under hard real-time constraints: lock-free audio threads, sub-millisecond latency, and Test-Driven Development (GoogleTest) across the build pipeline — the same concurrency, threading, and timing discipline required for safety-critical embedded firmware.
- Led real-time audio AI enablement R&D: trained deep models (PyTorch) to match target frequency responses for audio plugins, taking the approach from prototype to real-world deployment within the production DSP pipeline.

PhD Research — Embedded ML · Iowa State University

Jan 2021 – May 2025

Ames, IA · Edge-inference and analytics on top of the DigitalAg embedded sensor stack.

- Built CNN/Transformer pipelines (PyTorch) for sensor-to-image regression on CAN- and LiDAR-derived telemetry; improved field segmentation precision by 22%.
- Cut sensor-data preprocessing time ~30% on terabyte-scale datasets via scalable loaders, chunked ETL, and parallel I/O workers; published 3 peer-reviewed IEEE papers.

Data & Embedded Systems Intern · 3M · St. Paul, MN

May 2023 – Aug 2023

- Engineered a scalable cloud-based sensor-visualization platform on GCP (Python + SQL) ingesting real-time factory telemetry from embedded sensor nodes, enhancing operational insights and decision-making efficiency 30%.
- Improved airflow-sensor test efficiency 15% by designing a DOE-driven (Minitab) test methodology that eliminated redundant cycles at the hardware/software boundary; defined test-system architecture with cross-functional EE/ME/firmware engineers.

Engineer Designer II / Engineer I · Iowa State University · Ames, IA

Jan 2020 – Dec 2020

- Engineered a reusable Android field-documentation app with offline-first synchronization and structured metadata capture, improving field data-collection efficiency 40%.
- Developed automated SQL analytics and idempotent Python/Bash backup workflows, cutting data processing time 50% across multi-terabyte agricultural datasets.
- Executed MATLAB-based satellite imagery analysis (NDVI/EVI vegetation-index thresholding), achieving a 20% increase in crop loss prediction accuracy for scalable agricultural ML pipelines.

Advanced Manufacturing Engineering Co-Op · GE Appliances

Aug 2018 – Dec 2018

- Designed and deployed an embedded inventory-control system (barcode scanners over UART, MCU-driven label triggering, SQL reconciliation), improving tracking accuracy 20%.
- Reduced testing errors 15% by optimizing automated test sequences and camera-based vision-test rigs across the appliance production line.

Embedded Firmware Engineer · Heilbronn University, Germany

Feb 2017 – Mar 2017

- Delivered an ISO 15118-compliant display-driver firmware for a 4th-gen EV charging station (C++ / EBGuide HMI); led a 3-engineer team through driver development, hardware integration, CAN communication, and on-station debugging.

SELECTED EMBEDDED PROJECTS

- **GPS + CAN Field Tagger** — custom hardware + C firmware injecting GPS-timestamped events onto an equipment CAN bus; validated with oscilloscope and CAN analyzer.

- **Multi-Modal Perception Integration** — LiDAR + radar + stereo + vision cameras onto embedded controllers for field-vehicle autonomy: time sync, calibration, point-cloud processing, and fusion-ready output.
- **CAN / J1939 Data Extraction** — firmware + tooling to decode and log heavy-equipment bus traffic into structured datasets for analytics and ML.
- **Industrial High-Voltage Monitoring Node (ESP32-C6, 300VDC)** — protected voltage-divider front end (Zener clamp + TVS + ferrite), datasheet-driven component selection, oscilloscope validation, I2C/ADC firmware, telemetry to QuestDB + Grafana.
- **ROS Robotics (Iowa State Robotics Club)** — robot control, sensor topics, and perception nodes on the standard middleware used across modern autonomous stacks.
- **Self-Hosted Edge-Inference Cluster** — dual-GPU Linux server for containerized inference (Docker + NVIDIA Container Toolkit); resolved CUDA/driver conflicts and added auto-restart and observability.

EDUCATION

PhD, Computer Engineering · Iowa State University **2021 – 2025**
 GPA 3.82 · Focus: embedded ML, sensor data quality, augmentation, and pruning

MS, Computer Engineering · Iowa State University **2017 – 2019**
 GPA 3.92 · Specialization: Real-Time Systems + IoT · Capstone: custom sensor + MCU data-collection hardware

BS, Electrical Engineering · Anna University, India **2013 – 2017**
 Focus: embedded systems, power electronics, ISO 15118 EV-charging framework

SELECTED PUBLICATIONS

- IEEE Access, 2025 — Interpretable AI for Time-Series: Multi-Model Heatmap Fusion (94.1% accuracy on PhysioNet; RMSE 0.28 kWh on UCI Energy).
- IEEE ICECET 2025 (Paris) — Multivariate Temporal Regression at Scale: ML + XAI + NLP framework.
- IEEE ICMLA 2023 (Nassau) — Deep Learning and Pattern-Based Methodology for Multivariable Sensor Data Regression ($R^2 = 0.861$).